

PROJECT DESIGN PHASE-II

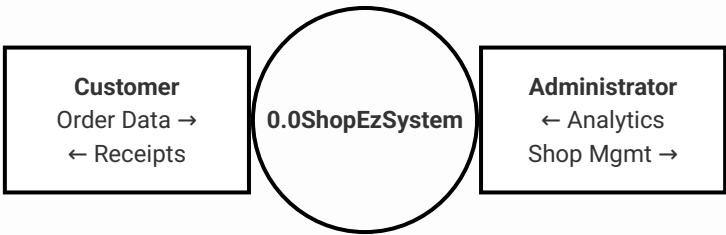
Data Flow Diagram & User Stories

Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN Application)
Maximum Marks	4 Marks

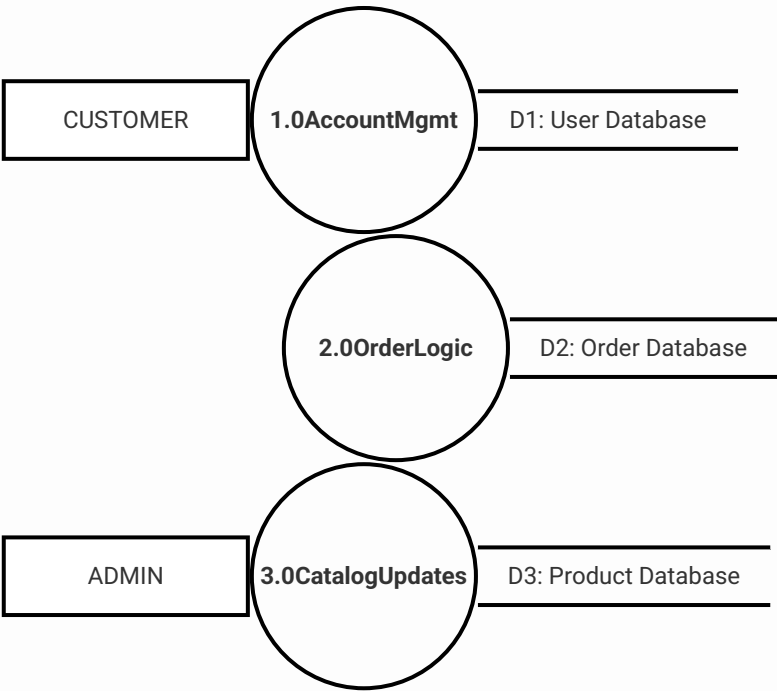
Data Flow Diagrams (DFD):

The schematics below visualize the movement and transformation of information within the ShopEz architecture, highlighting the primary interactions between end-users (Customer/Admin) and the underlying system processes.

Level 0: Context Diagram for ShopEz



Level 1: Process Decomposition



USER STORIES & TASKS

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Sprint
Customer	Account	USN-1	As a user, I want the ability to initialize a personal profile.	Verification of mail/pass formatting.	High	S1
Customer	Account	USN-2	As a user, I want to securely sign in to my existing account.	Validation via token-based session.	High	S1
Customer	Catalog	USN-3	As a shopper, I want to explore items grouped by specific categories.	Frontend UI filters populate product grid.	High	S2
Customer	Search	USN-4	As a shopper, I want to locate products using specific keywords.	Accurate keyword matches in item names.	Med	S2
Customer	Cart	USN-5	As a shopper, I want to save product selections to a digital cart.	Persistent basket state in DB store.	High	S3
Customer	Checkout	USN-6	As a buyer, I want to finalize and commit my shopping order.	Generation of an explicit order summary.	High	S3
Customer	History	USN-7	As a user, I want to review my previous purchase records.	Retrieval of sequential order history.	Med	S4
Admin	Inventory	USN-8	As a manager, I want to create or remove inventory listings.	Database records update instantaneously.	High	S4
Admin	Dashboard	USN-9	As a manager, I want to oversee all incoming client transactions.	Secure tabular view of all retail activity.	Med	S4